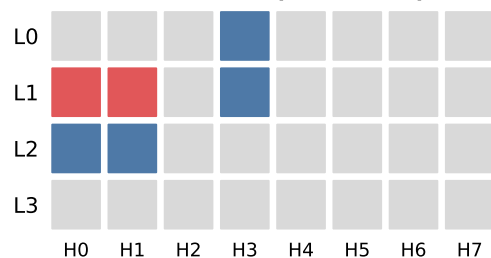


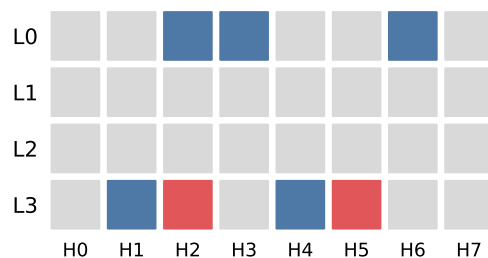
Phase 8 — Circuit Tracing: Trained vs Hardcoded SUBLEQ Transformer

Fetch Broadcast Other

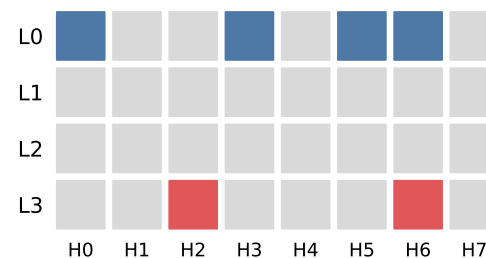
Hardcoded (reference)



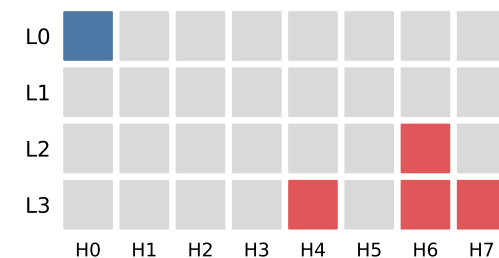
Trained — Seed 0



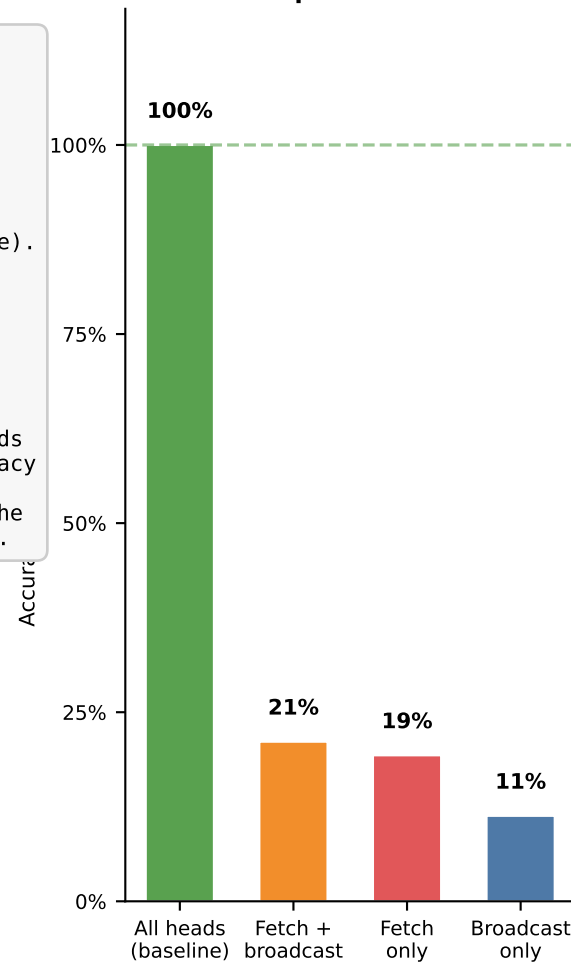
Trained — Seed 1



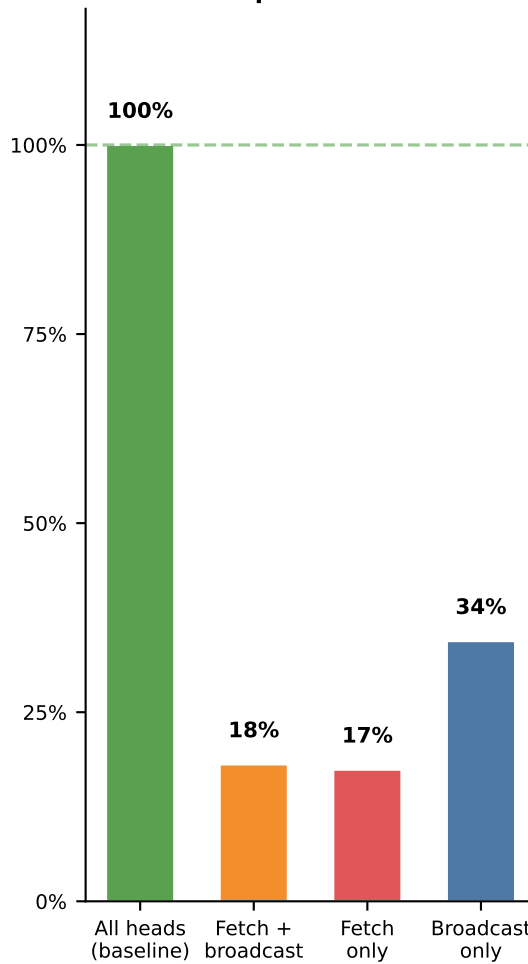
Trained — Seed 2



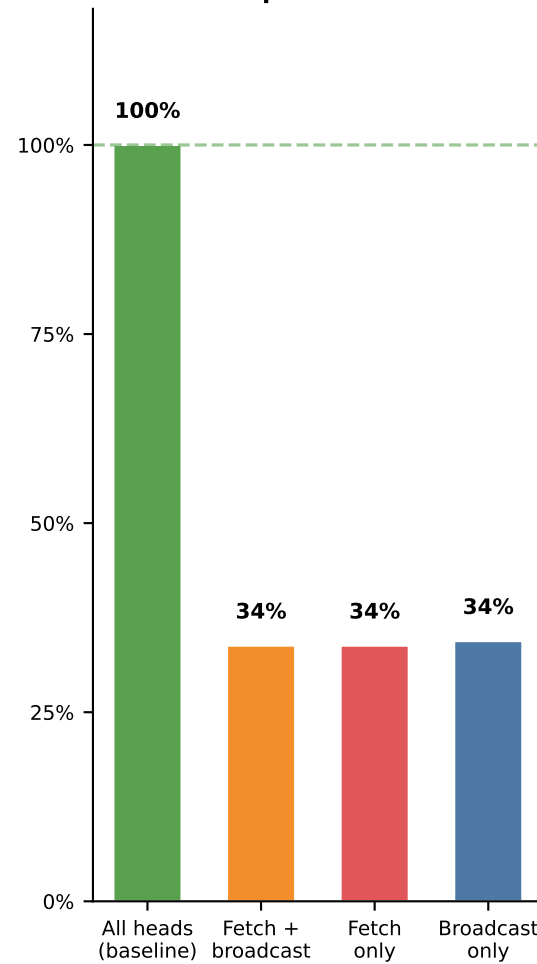
Circuit completeness — Seed 0



Circuit completeness — Seed 1



Circuit completeness — Seed 2



What the circuit tracing shows:

TOP ROW — Head roles

Each square is one attention head.
Red = fetch head (reads a specific memory address).

Blue = broadcast head (sends the current PC to all positions).

Grey = 'other' (no clear single role).

Hardcoded model has clean, sparse circuits: fetch in L1, broadcast in L0–L2. Trained models spread the same roles across L0 and L3.

BOTTOM — Circuit completeness

If we keep only fetch/broadcast heads and zero out everything else, accuracy drops to ≈20–34%. The 'grey' heads in L1–L2 are doing real work that the simple fetch/broadcast story misses.

Accuracy